

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
10	(a)	(i)		moles = $\frac{\text{conc} \times \text{volume}}{1000} = \frac{1.5 \times 12}{1000} = 0.018$ (2) accept 0.02 if working correct if answer incorrect award (1) for either of following 0.012 18 no ecf possible if formula is rearranged incorrectly		2		2	2	2
		(ii)		0.036 ecf possible from part (i)		1		1	1	1
		(iii)		3.6 (2) ecf possible from parts (i) and (ii) if answer incorrect award (1) for $\frac{0.036}{10}$		2		2	2	2
	(b)	(i)		award (1) for either of following any positive temperature change of less than 19°C temperature change will be less than 19°C / lower award (1) for reason e.g. (ethanoic acid) is a weaker acid / has a higher pH (ethanoic acid) is less dissociated / has fewer H ⁺ ions accept ethanoic acid / it is a weak acid			2	2		2

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					AO1	AO2	AO3	Total	Maths	Prac
		(ii)	I	copper(II) ethanoate accept copper ethanoate	1			1		
			II	$\text{Cu}(\text{CH}_3\text{COO})_2$			1	1		
				Question 10 total	1	5	3	9	5	7